

Saptarshi Mandal

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Education

University of Illinois Urbana-Champaign

PhD in ECE(Ongoing) ; Advisor - R. Srikant

GPA : 4/4

Aug 2021 - Dec 2026

Indian Institute of Technology, Kharagpur

Joint B.Tech-M.Tech, major in EECE and minor in CSE

GPA : 9.34/10

Aug 2016 - April 2021

Internships

Google Research

AI for Social Good

Supervisor - Prof. Milind Tambe, Harvard University; Collaborator: Niclas Boehmer, HPI, Germany

Oct - Dec, 2023

- Explored the use of data-driven approaches to efficiently allocate scarce resources and encourage patients to seek healthcare in rural India. Contributed to the modeling of general Restless Bandit framework suitable for the intervention effect dataset in Google Research, India .
- Accepted as a full paper at **International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2025**.
- Link to the paper: <https://arxiv.org/abs/2501.06103>.

NVIDIA

Bangalore, India

Project intern-digital profile

Summer, 2019

- Automated the fault aggregation part of error collation from the IP metadata of the Orin SoC architecture accessing the Google Drive API in the Pearl script and parsing IP metadata efficiently.
- Built a comprehensive latency calculator using the maps between IPs in the same SoC architecture. Exponentially reduced the latency calculation time using shortest path algorithms and recursion in Python.

Research

Distributionally Robust TD Learning and Q Learning with Linear Function Approximation

UIUC

Supervisors - : R. Srikant, UIUC

2024-2025

- We present the first robust TD learning and Q learning with linear function approximation, where robustness is measured with respect to the total-variation distance and Wasserstein-I distance uncertainty set without restrictive assumptions on the model. Our results close a key gap between the empirical success of robust RL algorithms and the non-asymptotic guarantees enjoyed by their non-robust counterparts.
- Under review at **International Conference on Learning Representations (ICLR) 2026**.
- Link to the paper : <https://arxiv.org/abs/2510.01721>.

Inferring Labels from Multi-type Crowdsourced Data

UIUC

Supervisor - Prof. R. Srikant, UIUC

2021 - 2022

- Extended the Dawid-Skene (DS) model to multi-type cases and analyzed the error bounds of existing algorithms. Focusing on the case where there are two types of tasks, we propose a spectral method to partition tasks into two groups that cluster tasks by type. Our analysis reveals that task types can be perfectly recovered if the number of workers scales logarithmically with the number of tasks.
- Accepted at **Transaction on Machine Learning Research (TMLR) 2025**.
- Accepted for a talk at **INFORMS Applied Probability Society Conference**.
- Link to the Paper: <https://arxiv.org/abs/2302.07393>.

A Theoretical Analysis of Soft-Label vs Hard-Label Training in Neural Networks

UIUC

Supervisors - : Prof. Lin Xiaojun, CUHK; Prof. R. Srikant, UIUC

2023 - 2024

- Theoretical insights for a few widely known phenomena of Knowledge Distillation including Model Compression (using traditional labels vs. soft labels for training in Supervised Classification task) using the tools of Neural Tangent Kernel.
- Accepted at the **Learning for Dynamics and Control (L4DC) 2025**.
- Link to the paper : <https://arxiv.org/abs/2412.09579>.

An Online Delay-reliable Efficient Algorithm for Multi-resource Allocation

IIT Kharagpur

Supervisor - Prof. Goutam Das, IIT Kharagpur

2020 - 2021

- Proposed a polynomial time delay reliable scheduling strategy for multi-resource allocation in wired media optimal in the presence of adversarial packet arrival; Proposed an approximate solution for multiple types of packets for the same.
- Manuscript under preparation.

Technical Skills

Programming Languages	Python, PyTorch, Matlab, C, C++, Verilog
Planning and Control	Robust RL, MuJoCo, Gym
Software and Platforms	Simulink, OMNeT++, CPLEX, Xilinx ISE, L ^A T _E X, Beamer, Git

Relevant Graduate Courses with Grades

Machine Learning	Stat. Learning Theory(A), Stat. Reinforcement Learning(A), Neural Networks and Applications(A)
Probability	Random Processes(A+), High Dimensional Prob.(A+), Information Theory (A+), Queueing Theory(A+)
Optimization, Game Theory	Adaptive Systems(A+), Network Optimization(A+), Game Theory(A+)
Algorithms	Approximation and Online Algorithms(A), Communication Signal Processing and Algorithms(A+)

Teaching Experience

UIUC	Graduate Course: “MDPs and Reinforcement Learning”	2025
UIUC	Graduate Course: “Introduction to Optimization”	2024
IIT Kharagpur	Under-Graduate Lab: “VLSI Lab”	2021
IIT Kharagpur	Under-Graduate Lab: “Microcontroller Systems Lab”	2020

Awards and Honors

IISc	Scholarship: “KVPY ”	2016
IISc	Fellowship: “Summer Research Fellow” by “Indian Academy of Science”	2020

Extra Academic Activity

- Former Hardware Modelling Team Member, LBS hall, IIT Kharagpur
- Former captain, Bengali Dramatics - Nehru Hall, IIT Kharagpur
- A Master of Ceremonies in the main event in Bengali Student Organization, UIUC